

3_data types Vs data structures explained

in C, C++ we need to declare `int a = 10`

here we are declaring data type explicitly as integer and value 10 is assigning to variable a

But python is dynamic programming language

so we are writing as `a = 10` in python

Purpose of data type is to determine how the data is stored in memory and what operations can be performed on top of data

Types of data types.

1. Text ==> strings --> "names", 'names', ""names""
2. Numeric ==> Integer, float, complex
3. Sequence ==> Lists, tuples and range functions
4. Mapping ==> Dictionaries
5. Set ==> sets
6. Boolean ==> True or False
7. Binary ==> 1 or 0
8. None ==> None

****Data Structures**

It is specialized format for organizing and storing data.

They provide and organize structured data so that it can efficiently processed and manipulated

purpose: organizing and managing large amounts of data, they facilitate various operations such as insertion, deletion, searching and traversal

In python, we have **Multiple data structures**

lists

tuples

dictionaries

sets

arrays and etc.

